Mini Data Science Project

Title: Student Performance Prediction Using Machine Learning

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1.Introduction

In education, data science and machine learning are increasingly significant, utilizing vast amounts of student data such as study time, attendance, and grades to analyze factors affecting academic performance and to predict future outcomes. Educators can identify students needing extra support and make data-informed decisions. A project employed the Student Performance Dataset to apply data science techniques, involving data preparation, exploration, visualization, and model evaluation to derive meaningful insights and forecast final grades.

1.1 Problem Formulation

Student academic performance is influenced by factors such as prior academic performance, attendance, and study habits. This project investigates the relationship between these academic variables and students' final grades, focusing on whether study time, absences, and prior grades can predict final outcomes.

The objectives that are discussed in this project are as follows:

1. To examine the relationship between study habits and absences and the performance of students.
2. To define which variables have the strongest correlation with final grades.
3. To forecast final grades of students using machine learning models.

Research Question

What is the relationship between study habits and academic factors and student performance and can machine learning models be precise predictors of student final grades?

1.2 Data Acquisition

The data set employed to do the project is the Student Performance Dataset that is available in the UCI machine learning repository. This data includes the information of the secondary school students and contains demographic, social, and academic characteristics.

In this analysis, the student-mat.csv file was taken. This file holds the student performance information of the subject mathematics.

Dataset Source

UCI Machine learning repository - Student performance dataset.

Dataset Description

The information contained in the dataset includes: School, Gender, Age, Address type, Family size, Parent education, Study time, Absences, First period grade (G1), Second period grade (G2), Final grade (G3)

Dataset Size

Number of rows is 395 and number of columns is 33. G3 is the target variable in this project as it is an end of course grade of every student.

2.Data Analysis

The dataset was cleaned and explored before implementing the machine learning methods, and it was necessary to learn the structure of this dataset and find the interesting patterns.

2.1 Data Cleaning and Preprocessing

The Pandas library was used to load the data into Python. The CSV file employed semicolon separators hence the sep=';' argument was used during the reading of the file. To ensure that the data was loaded successfully, the initial rows, columns names and the number of features were reviewed to confirm the dataset was loaded properly. It was then checked on missing values, duplicates and types of variables. There were no significant issues of missing values, and the data was appropriate to analyze. The machine learning task was chosen with the relevant academic variables, namely, studytime, absences, G1, G2, and G3. These data preprocessing operations were made to be ready to be explored and predicted.

2.2 Exploratory Data Analysis

To gain insight into the distribution of student performance and the relationship between significant variables, the Exploratory Data Analysis was performed. There are five visualizations that were produced to answer the research question.

Figure 1: Distribution of Final Grades

The histogram of end grades provides the general distribution of student grades. The majority of students rated within the middle range with the smaller number receiving either very low marks or very high marks. This means that there is a moderate distribution of students in their performance with most of them performing at an average performance.

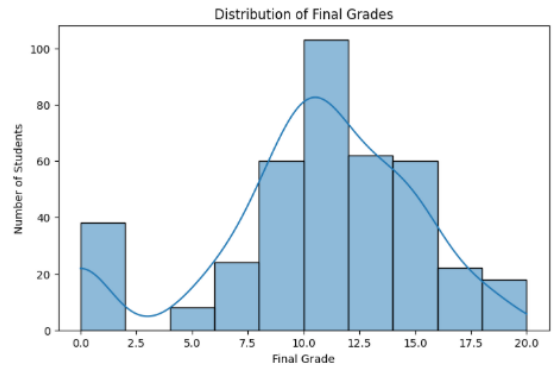

Figure 1:Distribution of Final Grades

Figure 2: Final Grade vs Time of Study

The scatter plot of study time and the final grade demonstrates that there is a positive relationship between the variables. Students that took more time to study usually tended to record higher final grades. Even though it is not a linear relationship, it is possible to say that the study habits are significant factors of academic achievement based on the graph.

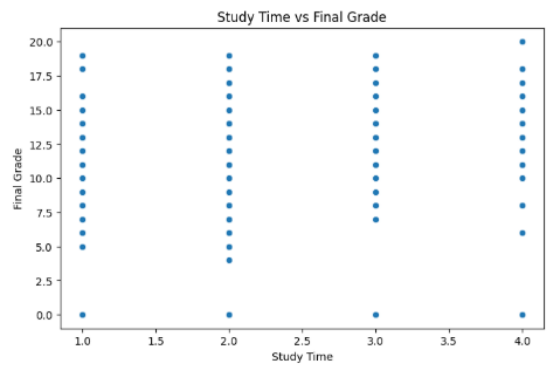

Figure 2:Final Grade vs Time of Study

Figure 3: absenteeism and final grade

The comparison graph indicates a correlation between student absences and final grades, revealing that higher absence rates are linked to lower final grades. This suggests that school attendance significantly influences student performance, as regular attendance generally promotes better academic outcomes.

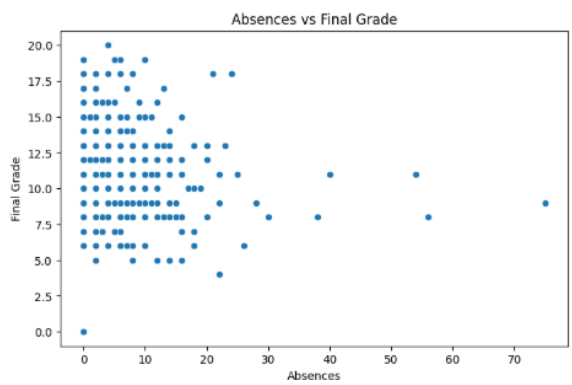

Figure 3:absenteeism and final grades

Figure 4: Heatmap Correlation

The relationship between the numeric variables in the dataset was measured with the help of the correlation heatmap. The best positive correlations with the final grade (G3) were determined with the previous grades G1 and G2. This means that previous academic performance is a good predictor of ultimate academic outcomes. Other variables like the time of study indicated less significant but significant relationships.

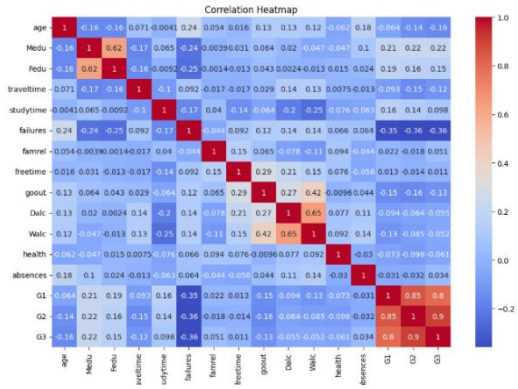

Figure 4:: Correlation Heatmap of Numerical Variables

Figure 5: Study Time Category Final Grade

The box plot used to compare the categories of study time and the final grades aid in generalizing the study results in terms of performance variations among various study habits. Higher categories of students in study time tended to perform better in median performance than the lower categories of students. It is corroborated by the hypothesis that higher levels of study effort are added value to academic performance.

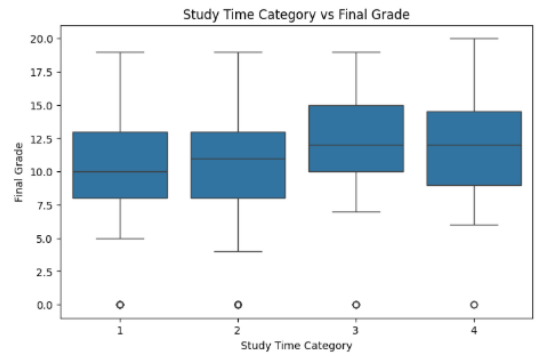

Figure 5:: Study Time Category Final Grade

Overall, the exploratory analysis revealed a moderate correlation between parental affection and child-parent attachment, as well as between child-parent attachment and identified psychopathological symptoms.

3.ML / AI Methods

Once the exploratory was done, machine learning models were used to predict the final grades of the students. The target variable G3 is a numerical variable and hence regression-related methods were chosen.

Predictor Variables

These were the features in the form of the following variables: Studytime, absences, G1, G2

Target Variable

- G3 (Final Grade)

The data was divided into 70 percent training and 30 percent testing data with the random state of 42. The models were built on the training data and the performance of prediction assessed on the testing data.

The following three regression models were applied in this project:

1. Linear Regression
2. Random Forest Regressor
3. Decision Tree Regressor

3.1 Linear Regression

The baseline model for predicting student grades was linear regression, which is a simple and interpretable method. The model achieved an R^2 score of 0.81, indicating it explains 81% of the variation in final grades, with a mean absolute error (MAE) of 1.30 and a mean squared error (MSE) of 4.25, demonstrating its effectiveness as a predictor.

3.2 Random Forest Regressor

An ensemble learning model, the Random Forest Regressor, was utilized due to its ability to integrate multiple decision trees, allowing for more complex data relationships than simple linear models. It achieved an R2 Score of 0.87, MAE of 1.05, and MSE of 2.90, outperforming all other models tested in predicting student final grades.

3.3 Decision Tree Regressor

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3.4 Model Performance Comparison

The following table summarizes the performance of all three models:

Model	R^2 Score	MAE	MSE
Linear Regression	0.81	1.30	4.25
Random Forest Regressor	0.87	1.05	2.90
Decision Tree Regressor	0.83	1.16	3.85

The comparison clearly shows that Random Forest Regressor was the best-performing model.

4.Results and Interpretation

The project demonstrates that machine learning and data science may be applied to analyse and forecast student performance. It was discovered that previous grades, absences, and study time have an extreme influence on the final grade. Random Forest provided the best results and the highest accuracy and minimum error were achieved. On the whole, the project demonstrates that machine learning can be used in comprehending the performance of students and making more advantageous academic choices.

5.Reflection

This project provided me with experience in the entire process of data science, including data collection, cleaning, visualization and machine learning. The difficulty of loading the dataset correctly in Python and selecting the most useful features to make predictions was one of them. The other problem was how to make proper graphs and be able to explain them. This assignment has taught me that information cleaning is extremely crucial prior to implementing machine learning models. I also got to know that visualizations can be used to obtain patterns and aid in a better analysis. Altogether, the given project enhanced my knowledge of Python and demonstrated that data science can be applied to obtain useful insights using actual educational data.

7.Reference

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